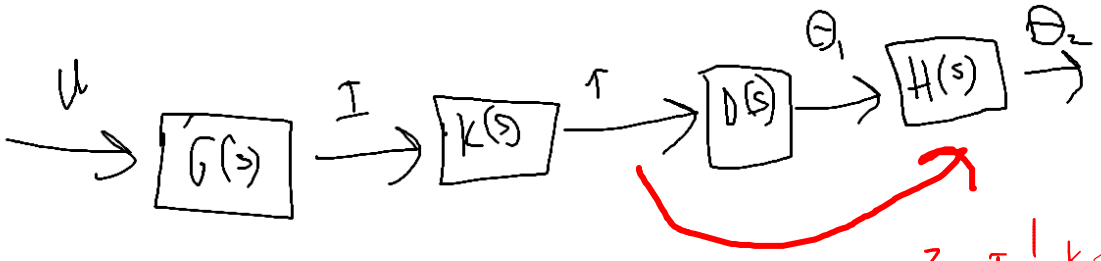
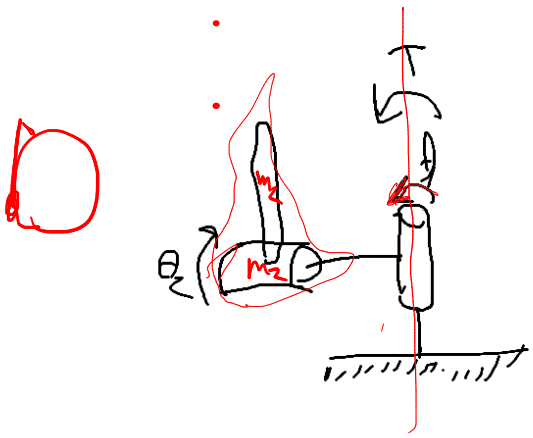




$$E_{kin} = \omega_1^2 \cdot I_1 \cdot \frac{1}{2} + \cancel{\omega_1^2 \cdot I_{z,1}} \cdot \frac{1}{2} + \omega_2^2 \cdot I_2 \cdot \frac{1}{2}$$

$$E_{kin} = \frac{1}{2} \cdot \left(\omega_1^2 \cdot (I_1 + I_{z,1}) + \omega_2^2 \cdot I_2\right)$$

$$E_{pot} = \sin(\theta_2) \cdot m_2 \cdot g \cdot L$$

$$\frac{d}{dt}\frac{dL}{d\dot{q}} - \frac{dL}{dq} = Q$$

$$L = E_{kin} - E_{pot}$$

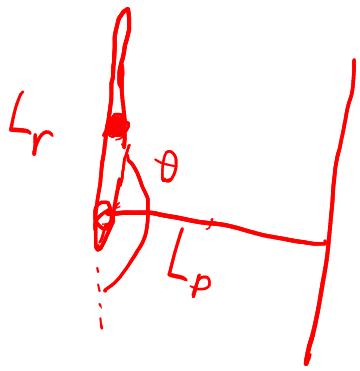
$$\boxed{q = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}}$$

$$\frac{dL}{dq} = \begin{pmatrix} 0 \\ \cos(\theta_2) \cdot m_2 \cdot g \end{pmatrix}$$

$$E_{pot} = \sin(\theta) \cdot m \cdot g$$

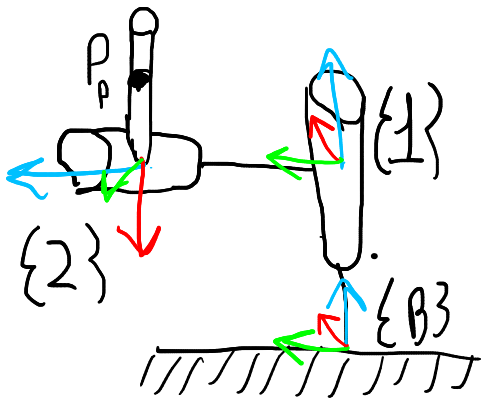
$$E_{pot} = (1 - \cos(\theta)) \cdot m \cdot g \cdot l$$

$$I_{z,1} = \left(L_r^2 + \frac{L_r^2}{4} \sin(\theta)^2\right) \cdot m$$



tavle 2

Her er theta 1 90 grader



$$ {}^{B}_{1}T = \begin{pmatrix} R_z(\theta_1) & 0 \\ 0 & 1 \end{pmatrix} $$

$$ {}^{1}_{2}T = \begin{pmatrix} R_x(-90^\circ) \cdot R_z(90^\circ + \theta_2) & \begin{pmatrix} L_r \\ 0 \\ 0 \end{pmatrix} \\ 0 & 1 \end{pmatrix} $$

$$ {}^{B}_{2}T = {}^{B}_{1}T \cdot {}^{1}_{2}T $$

$$ {}^{B}P_p = {}^{B}_{2}T \cdot {}^{2}P $$

$$ {}^{2}P_p = \begin{pmatrix} L_p \cdot \frac{1}{2} \\ 0 \\ 0 \end{pmatrix} $$

$$ \boxed{ {}^{B}P_{2\,org} = {}^{B}_{2}T \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} } $$

$$ {}^{B}P_{2\,org} = \begin{pmatrix} L_r \cos(\theta_1) \\ L_r \sin(\theta_1) \\ 0 \end{pmatrix} $$

$$ {}^{B}P_p = {}^{B}P_{2\,org} + \begin{pmatrix} L_p \cos(\theta_1) \cos(\theta_2 + 90) \\ L_p \sin(\theta_1) \cos(\theta_2 + 90) \\ -L_p \sin(\theta_2 + 90) \cdot \frac{1}{2} \end{pmatrix} $$

$$V_p^2 = \frac{L_p^2 \dot{\theta}_1^2}{8} + \frac{L_p^2 \dot{\theta}_2^2}{4} + L_r^2 \dot{\theta}_1^2 + \frac{L_p^2 \cos(2\theta_2 + 180)\dot{\theta}_1^2}{8}$$

$$+ L_p L_r \cos(\theta_2 + 90)\dot{\theta}_1^2$$

$$V_p^2 = L_p^2 \cdot \left(\frac{1}{8} \cdot \dot{\theta}_1^2 + \frac{1}{4} \cdot \dot{\theta}_2^2 - \cos(2\theta_2)\dot{\theta}_1^2 \cdot \frac{1}{8}\right.$$

$$\left. - L_r \cdot \sin(2\theta_2)\dot{\theta}_1^2\right) + L_r^2 \dot{\theta}_1^2$$